

- 1 Some students researched how yeast, *Saccharomyces cerevisiae*, is used in the food industry. The students found that in different manufacturing processes yeast is provided with different types of sugar. These sugars are either monosaccharides or disaccharides.

The students decided to compare the effect of supplying yeast with glucose or with maltose.

Glucose is a monosaccharide. Maltose is a disaccharide composed of two glucose molecules joined by a glycosidic bond.

- The students made a standard yeast suspension using 100g of dried yeast and 1 dm³ of deionised water.
- The suspension was left at 20 °C for two hours. During this time the yeast became active.
- After two hours the yeast culture was divided into two samples.
- Sample **A** was provided with glucose and sample **B** was provided with maltose.
- The apparatus shown in Fig. 1.1 was set up for each sample separately.

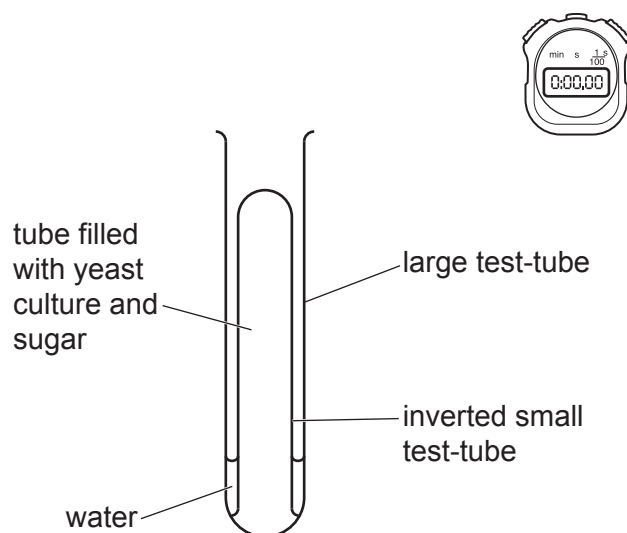


Fig. 1.1

- As the yeast respired, gas collected at the top of the small test-tubes, as shown in Fig. 1.2.

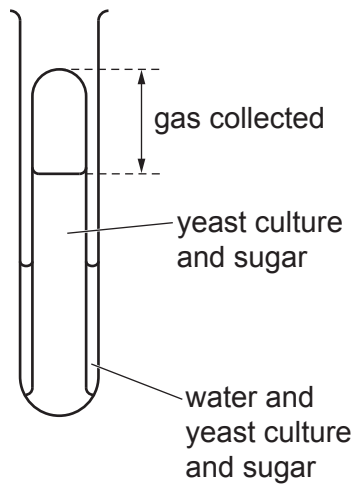


Fig. 1.2

The students used the apparatus to test the hypothesis:

Yeast has a higher rate of respiration when supplied with glucose than when supplied with maltose.

(a) (i) State the independent variable **and** the dependent variable in this investigation.

independent variable

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dependent variable

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[2]

(ii) The students stirred the yeast suspension before dividing it into the two samples.

Explain why it is important to stir the suspension.

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Question 1 continues on page 6.

- (b) Other students used a gas syringe to measure the volume of gas produced when a suspension of yeast was supplied with glucose or maltose.

Fig. 1.3 shows their results.

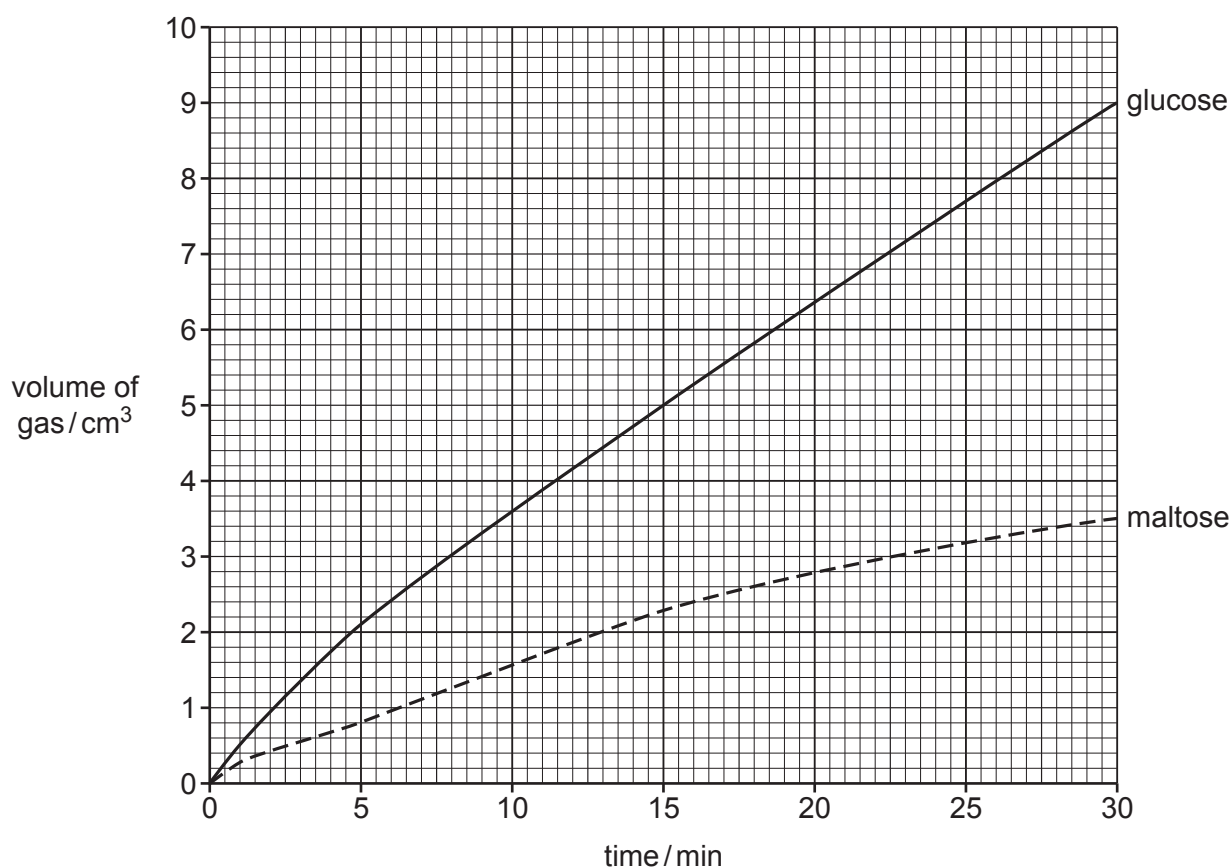


Fig. 1.3

- (i) State how the students could calculate the rate of respiration from the results shown in Fig. 1.3.

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(ii) State **and** explain the conclusions that can be made for the results in Fig. 1.3.

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(c) When the students evaluated their method they thought that they could measure the change in pH in the yeast suspension as the gas produced by yeast is carbon dioxide.

One student suggested using an indicator solution and colour chart. Another student suggested using a pH meter.

Suggest the advantages of using a pH meter rather than an indicator solution and colour chart.

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[Total: 17]